■ Multi-Agent AI System Notes

This document summarizes the 14 key AI technologies of 2025 and shows how they fit together inside a Multi-Agent System architecture. The technologies are divided into three layers: Core Agents, System Intelligence, and Infrastructure & Governance.

■ Core Agents (Specialists)

- Generative AI / LLMs → Create text, images, and code (e.g., ChatGPT, MidJourney).
- Multimodal AI → Understands text, images, audio, and video together (e.g., GPT-4o, Gemini).
- Vision-Language-Action (VLA) → Sees, understands, and acts (e.g., robotics models like Helix, GR00T).
- Domain-Specific AI → Specialized expert AIs for medicine, law, finance (e.g., Med-PaLM, BloombergGPT).

■ System Intelligence (Coordinators)

- Agentic AI → Plans and works on tasks independently (e.g., AutoGPT, Copilot agents).
- Multi-Agent Systems → Multiple AIs collaborate like a team (e.g., Stanford Smallville, AutoGen).
- Embodied AI → AI in a physical body like a robot, car, or drone (e.g., Tesla Optimus, Boston Dynamics).

■ Infrastructure & Governance (Support)

- Edge / On-Device AI → Runs locally on phones/robots without internet (e.g., Apple Intelligence, Pixel Nano).
- Long-Context Models → Handle very long memory (e.g., Claude 3.5, Gemini 1.5).
- AI Infrastructure → GPUs, TPUs, and supercomputers powering AI (e.g., NVIDIA Blackwell, Google TPUs).
- Explainable & Trustworthy AI → Shows reasons for AI decisions (e.g., IBM OpenScale, EU AI Act).
- Sustainable / Green AI → Energy-efficient AI (e.g., Google carbon-neutral AI).
- Hyper-Personalization & Ambient AI → AI adapts to your life, runs in background (e.g., Spotify AI, Apple).
- AI-Robotics Integration → Smarter robots powered by AI (e.g., Amazon robots, Agility Digit).

■ Example: Autonomous Cars

Autonomous cars are a combination of many AI technologies:

- Core: Embodied AI + VLA + Edge AI (car sees, plans, and acts in real time).
- System Intelligence: Agentic AI for route planning, Multi-Agent Systems for car-to-car coordination.
- Infrastructure: Long-context memory, Explainable AI (safety), and domain-specific driving AI.